Reg. No. | | | | | | | | | | [14AE32]

THIRD SEMESTER B.TECH. EXAMINATION, DECEMBER – 2015
MEASUREMENTS AND INSTRUMENTATION FOR AEROSPACE

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full questions taking atleast two questions from each part.
2. Draw clean sketch wherever required.

PART - A

1.	**a.**	Define accuracy, precision, reproducibility and repeatability as applicable to cockpit instruments.	**07**
	b.	State the four principle mechanical elements that constitute an instrument and explain the function of each element.	**07**
2.	**a.**	Name some of the aircraft instrument to which a digital counter display is applied.	**02**
	b.	What is the significance of colored markings applied to the dials of certain instruments with a neat sketch?	**06**
	c.	What are qualitative and quantitative instruments in a cockpit? Explain with examples.	**06**
3.		Draw a schematic diagram of complete pitot static system with sensor, tubing and cockpit indicators and explain how the entire system works.	**14**
4.	**a.**	What is a Gyroscope and what are its major applications in the field of aerospace?	**07**
	b.	What are the examples of gyroscope which work on different operating principles other than rotating fly wheel? Briefly explain.	**07**

PART - B

5.	**a.**	List the systems, location in aircraft, helicopter where the measurement of rpm assumes greater importance. Explain the systems with the need for rpm measurements.	**07**
	b.	The gas generator rpm of a gas turbine of an engine in a helicopter is 7200. The signal output frequency is 240Hz. What is the number of sensors on the shaft (PPR).	**07**
6.	**a.**	How is a current limiting resister is employed in a Wheatstone bridge and what is the purpose of the same?	**06**
	b.	A Wheatstone bridge is employed in the measurement of outside air temperature	**08**

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with a thermal sensing element having 100 ohms at $0^{\circ}C$; the three dummy resistors are having precision values of 100 ohms each. If the excitation voltage is 5 volts, Calculate the power dissipated in the sensor at $0^{\circ}C$, if the excitation is changed to 2 volts, what is the power dissipated at $0^{\circ}C$?

7.	**a.**	What is the principle of operation of fuel quantity measurement?	**06**
	b.	Explain the operation of capacitance based fuel quantity measurement with a schematic diagram.	**08**
8.	**a.**	How does a PCM system works in a digital data acquisition of FDR?	**07**
	b.	An FDR in a helicopter employs digital PCM recording with bit rate of 12KB/Sec. The recording consists of sequence of single frame. The frame consists of 64 words (including two sync words) of 12 bit each. Following parameters occur in the following rates in each frame: **i.** Altitude: Once in a frame **ii.** Longitudinal Acceleration: Four times in a frame **iii.** Pitch rate: twice in a frame **iv.** Lateral cyclic pitch: Twice in a frame What are the sampling rates of each of the above parameters?	**07**